

COP509: Natural Language Processing

Coursework: Document Analysis System

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Weighting: 100%

Deadline: 20 May 2026, 2pm

Submission: Learn

Overview

Government bodies and public organisations routinely publish inquiry reports with recommendations and accompanying response documents outlining their position on each recommendation. Manually tracking which responses address which recommendations is time-consuming and error-prone. Your task is to develop a document analysis system that enables users to (1) search a document collection, (2) extract recommendations from the recommendation documents, (3) identify responses from the response documents, and (4) align recommendations with their corresponding responses.

Dataset

You are provided with six document pairs via Learn (and the module's Github page). Each pair consists of a recommendation document and a corresponding response document.

Task 1: Search system

(15 marks)

Build a search system that allows users to find relevant passages within the document collection. Consider how to handle PDF text extraction and how to segment documents into searchable units. Develop a methodology for evaluating your search system, document your approach, and present the results.

Task 2: Extraction and Alignment

(65 marks)

Task 2a: Recommendation Extraction (35 marks)

Build a system that identifies and extracts recommendations from the recommendation documents. Recommendations appear in various forms: explicitly numbered, as statements using words like “should” or “must”, or embedded within paragraphs. Assign confidence scores reflecting how certain your system is about each extraction. Distinguish actual recommendations from text that only references or discusses them. Output extractions in a structured table format.

Task 2b: Response Matching and Classification (25 marks)

For each recommendation, find and extract the corresponding response from the response document. Some recommendations may not have a response. This task is challenging because responses may not directly quote the recommendation; they may paraphrase, summarise, or address multiple recommendations together. Assign confidence scores to each alignment indicating how certain your system is about the match. Classify each response as accepted, partially accepted, rejected, or not addressed.

Task 2c: Evaluation (5 marks)

Present a table showing the ground truth alignments alongside your system's extracted alignments, and the expected vs actual outcomes (accepted, partially accepted, rejected, not addressed). A sample table format is provided on Learn. For each row, provide a comment indicating whether it matches, any issues identified, and for errors, explain why it occurred and suggest how it could be fixed.

Interactive Demos

(10 marks)

Include interactive interfaces using ipywidgets to demonstrate your system's functionality for search, extraction, and alignment. Interfaces should be intuitive and allow users to explore the system's behaviour.

Documentation and Presentation

(10 marks)

Your notebooks must be written in tutorial style, well-documented and clearly presented in the form of a tutorial. See [this example tutorial](#) for the expected style.

Additional Information

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- **Tackling challenges:** You are expected to encounter several challenges during development. These must be clearly labelled in your notebook, together with an explanation of how each challenge was addressed. You must provide evidence demonstrating that the proposed solution resolves or mitigates the issue. Failure to document challenges and resolutions will negatively impact marks. **Note:** You are not expected to solve all challenges—focus on demonstrating depth over breadth.
- **Evaluations:** Keep track of all your evaluations and code versions during development. In your notebook, present only the results of your final evaluation.
- **Feedback:** Please ask for feedback during the lab sessions.
- **Packages:** Use any Python packages you see fit for your implementation.

Submission

Submit a ZIP file containing:

1. COP509_Notebook1_Search.ipynb — covers Task 1
2. COP509_Notebook2_Extraction_Alignment.ipynb — covers Task 2 (2a, 2b, 2c)
3. COP509_Notebook1_Search.pdf — PDF version of Notebook 1
4. COP509_Notebook2_Extraction_Alignment.pdf — PDF version of Notebook 2

Note: You may submit one notebook (.ipynb file and its corresponding PDF) with both tasks if you prefer.

Converting to PDF: Run the following in a Colab cell to generate a formatted PDF (example code below):

```
!pip install nbconvert
!apt-get install texlive-xetex texlive-fonts-recommended texlive-latex-extra -y -qq
!jupyter nbconvert --to pdf "your_notebook.ipynb"
```

Note: Replace "your_notebook.ipynb" with your actual notebook filename, e.g., "COP509_Notebook1_Search.ipynb".

Marking Criteria

Use this rubric to self-assess your work before submission.

Criterion	Max	Distinction (70–100%)	Merit (60–69%)	Pass (50–59%)	Fail (0–49%)	Mark
Task 1: Document Search (15 marks)						
Text extraction and preprocessing	5	Text extracted cleanly from all PDFs; appropriate preprocessing applied	Extracts most text; minor issues with some documents	Basic extraction with some errors; limited preprocessing	Fails to extract or significant errors in text quality	
Search implementation	5	Returns relevant results ranked appropriately; handles various queries	Good search with mostly relevant results	Basic search returns some relevant results	Search fails or returns irrelevant results	
Evaluation methodology and results	5	Clear methodology explained and justified; appropriate metrics chosen; thorough analysis	Good methodology; results presented clearly with some analysis	Basic methodology described; some results shown	No methodology or unclear approach; results missing	
Task 2a: Recommendation Extraction (35 marks)						
Extraction accuracy	15	Correctly identifies recommendations with few false positives/negatives	Most recommendations extracted; minor errors	Extracts some recommendations; several errors	Misses many recommendations or many false extractions	
Confidence scoring	5	Scores reflect extraction certainty; higher scores for clear recommendations	Good scores with reasonable justification	Basic scores assigned	No scores or arbitrary values	
Handling different formats	10	Handles numbered, modal verb, and embedded recommendations	Handles at least two formats well	Handles at least one format	Only partially handles one format	
Solutions to other challenges	5	Solution implemented and effectively resolves the challenge	Good solution that mostly addresses the challenge	Solution implemented but does not fully solve the challenge	Solution does not work or no attempt	
Task 2b: Response Matching and Classification (25 marks)						
Recommendation-response matching	15	Correctly matches most recommendations to their responses; demonstrates sophisticated approach to handle difficult cases (e.g., paraphrasing, synonyms, semantic similarity)	Good matching accuracy; addresses key challenges effectively	Basic matching with reasonable accuracy; attempts to address some challenges	Poor matching accuracy or superficial approach with many errors	
Alignment confidence scores	3	Meaningful scores reflecting match certainty	Good scores with justification	Basic scores assigned	No scores or arbitrary values	
Response classification	7	Correctly labels as accepted/rejected/partial/not addressed	Most labels correct; minor errors	Some labels correct; several errors	Labels missing or mostly incorrect	
Task 2c: Evaluation (5 marks)						
Expected vs actual table with error analysis	5	Clear table showing ground truth vs system output; errors explained with suggested fixes	Good comparison; most errors analysed	Basic comparison shown; limited analysis	No comparison table or error analysis	
Interactive Demos (10 marks)						
ipywidgets interfaces	10	Functional, polished interfaces for all tasks (search, extraction, alignment); intuitive to use; enhances understanding of system behaviour	Good interfaces for most tasks; minor usability issues	Basic interfaces present but limited functionality or coverage	No interfaces or non-functional	
Documentation and Presentation (10 marks)						
Tutorial-style documentation	10	Notebooks read as a clear, step-by-step tutorial that someone else could follow and learn from; reasoning and decisions well justified throughout; code clearly commented	Good documentation with clear explanations; most decisions justified; generally easy to follow	Basic explanations present; some gaps in reasoning; partially followable	Little or no documentation; difficult to follow; reasoning unclear or missing	
Total						/100

Academic integrity: Standard NLP libraries and pre-trained models are permitted. Cite all external resources.